MIL-A- 47179 (MI)
7 June 1974
SUPERSEDING
MIS-11034
14 September 1964

Military Specification

ASBESTOS FIBRE, CHRYSOTILE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of chrysotile asbestos fibre.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9390

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60605.)

National Motor Freight Traffic Associations, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1616 P Street, N.W., Washington, D.C. 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When required (see 6.2) a preproduction sample shall meet the requirements of this specification.

3.2 Storage. The chrysotile asbestos fibre shall meet the requirements of this specification after storage at 32 degrees Celsius (C) (90 degrees Fahrenheit (F)) maximum for a period of one year minimum.

3.3 Chemical and physical. Chemical and physical requirements of the material shall be in accordance with Table I.

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TABLE I

Chemical and Physical Requirements

Characteristic	Requirement	
	Minimum	Maximum
pH Value	9.5	10.5
Heat Loss @ 1038 degrees C (1900 degrees F) percent	12.0	15.0
Water Solubles, percent		0.5
SiO_2, percent	38.0	42.0
MgO, percent	40.0	42.0

3.4 Workmanship. The chrysotile asbestos fibre furnished under this specification shall be a product free from foreign materials. It shall be uniform in quality and manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibiliby for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. A lot shall consist of material produced by one manufacturer in one continuous production period with no change in process or formulation, or manufacturer's brand of raw ingredients.

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4.3 Quality conformance sampling. Sampling by the purchaser shall be in accordance with variables sampling plans as specified in MIL-STD-414, using the standard deviation method, variability unknown. Double specification limit with an Acceptable Quality Level of 1.5 shall be used. An inspection lot represented by a container of material not conforming to the requirements of this specification will be subject to rejection.

4.4 Classification of inspection. The inspection and testing of chrysotile asbestos fibre shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.5 Preproduction tests. Preproduction tests shall consist of all the tests specified in 4.7.

4.6 Quality conformance tests. Quality conformance tests shall consist of pH value and heat loss as specified in 4.7.1 and 4.7.2.

4.7 Test methods.

4.7.1 pH value.

4.7.1.1 Apparatus.

a. Beckman Zeromatic pH meter, or equivalent.

4.7.1.2 Reagents.

a. Standard buffer solution.

4.7.1.3 Procedure. The test shall be performed in accordance with the following steps:

a. Add 50 milliliter of freshly boiled distilled water adjusted to pH7 to 20 grams of the fibre sample.

b. Stir the solution well and cool to room temperature.

c. Clean the electrodes of a Beckman Zeromatic pH meter and adjust the instrument either by a pH meter tester or by standard buffer solution. Wash the electrodes with distilled water and wipe them dry with paper tissue.

d. Place the electrodes in the solution and determine the pH. Report the pH of solution to the nearest 0.1 pH unit.

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4.7.2 Heat loss, percent.

4.7.2.1 Apparatus.

a. Platinum crucible, or equal.

b. Muffle furnace.

c. Desiccator.

4.7.2.2 Procedure. The test shall be performed in accordance with the following steps:

a. Weigh a 1 gram sample of material into a tared, ignited, platinum crucible.

b. Dry 1 hour at 104 degrees C (220 degrees F).

c. Place in a muffle furnace at 1038 degrees C (1900 degrees F) and ignite for one (1) hour.

d. Cool in a desiccator and weigh.

e. Calculate as follows:

$$\text{Percent Heat Loss} = \frac{(\text{Total wt before ignition} - \text{total wt after ignition})}{\text{Wt before ignition}} \times 100$$

4.7.3 Water solubles, percent.

4.7.3.1 Appartus.

a. 250 milliliter beaker.

b. 250 milliliter volumetric flask.

c. Pipette.

d. Evaporation dish.

e. Desiccator.

4.7.3.2 Procedure. The test shall be performed in accordance with the following steps.

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a. Weigh approximately 2.5 grams of dry sample to the nearest 0.1 milligram and place in a 250 milliliter beaker.

b. Add 100 milliliters of distilled water and boil for five minutes.

c. Allow to cool, transfer to a 250 milliliter volumetric flask, and fill to mark with distilled water.

d. Allow to settle and transfer with a pipette, 50 milliliters of the clear solution to a weighed evaporation dish.

e. Evaporate to dryness and heat for one hour at 104 degrees C (220 degrees F) plus or minus 1 degree C (2 degrees F).

f. Place the dish in a desiccator, cool for approximately 20 minutes and weigh.

g. Calculate the percent water soluble material as follows:

$$\text{Percent Water Soluble} = \frac{(R - T)\ 500}{W}$$

Where: R = Weight of dish and residue in grams

T = Weight of dish in grams

W = Weight of sample in grams

4.7.4 Silicon dioxide, percent.

4.7.4.1 Apparatus.

a. Platinum crucible, or equal.

b. Meker burner, or equal.

c. 300 milliliter porcelain casserole, or equal.

4.7.4.2 Reagents.

a. Na_2CO_3, 2 normal.

b. HCl, concentrated.

c. MgO, pure.

d. H_2SO_4, concentrated.

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4.7.4.3 Procedure. The test shall be performed in accordance with the following steps:

a. Heat the tared, covered, platinum crucible containing the intimate mix of powdered specimen and Na_2CO_3 to drive out any moisture present. Gradually raise the temperature until the highest heat of a good Meker burner is obtained. As soon as the mass melts and there is not further evolution of carbon dioxide, the decomposition is complete.

b. Remove the crucible from the flame and carefully roll it at an angle so that the molten material makes a layer on the wall of the crucible. Allow the material to cool and place the crucible in a 300-milliliter poreclain casserole. Add 25 milliliters of water, partially cover the casserole, and carefully add 25 milliliters of 1:1 HCl.

c. A lively evolution of carbon dioxide takes place immediately. but as the silicic acid separates, the inner part of the cake gradually becomes coated with a film of silicic acid which protects it from further action of the acid. Consequently, it is necessary to break up the cake from time to time by means of a glass rod, until there is no evolution of gas and no more hard lumps remain.

d. After the evolution of carbon dioxide has nearly ceased, remove the watch glass and wash it with water. Remove the crucible from the casserole and carefully wash it with water, collecting the wash water in the casserole. Recover the casserole with the watch glass and evaporate to dryness. Heat the residue for at least an hour at 110 degrees C (230 degrees F) to dehydrate the silica.

e. Moisten the dry powder with concentrated hydrochloric acid and allow the covered dish to stand 10 minutes at ambient temperature to allow the basic salts and oxides formed during the evaporation and drying, to be changed to chlorides. Warm gently, dilute with 100 milliliter water, heat to boiling and after the silicic acid has settled, filter through a well-fitted ashless filter paper. Wash the residue three or four times by decantation with 2 N hydrochloric acid. Transfer to the filter and wash with hot water until free from chlorid. Place the precipitate in a platinum crucible and set aside.

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f. The separation of the silicic acid is not quite complete; as much as 2 milligrams may remain in the filtrate. To remove this, evaporate the solution to dryness on the water bath, and again heat at 110 degrees C (230 degrees F) for an hour. Moisten the residue with 5 milliliter of concentrated hydrochloric acid, and allow to stand not more than 15 minutes. Warm, dilute to 100 milliliter, heat to gentle boiling, and filter through a new and correspondingly small, ashless filter, washing with hot 2 N acid and with water as in step (e). Save the filtrate and analyze it for MgO per 4.7.5.

g. Combine the second wet filter with the one previously obtained and ignite the wet filters containing the silica in a tared, platinum crucilbe. Keep the temperature low until all carbon is consumed. Do not allow the filter to catch fire. Cover the crucible, ignite over a Meker burner and cool. Weigh the crucible and the residue (A).

h. Moisten the residue with water, add 3 drops of concentrated sulfuric acid and about 20 milliliter of pure hydrofluoric acid. (do not measure this acid in a glass graduate. Do not inhale the fumes. If spilled on the hands, flush immediately with water.)

i. Place the crucible in an air bath and evaporate under a hood until no more vapors are expelled. Remove the excess sulfuric acid by heating over a free flame. Raise the temperatures gradually and finally heat the crucible over a Meker burner, cool and weigh (B).

j. Calculate as follows:

$$\text{Percent } SiO_2 = \frac{A - B}{\text{sample wt}} \times 100$$

4.7.5 Magnesium oxide, percent.

4.7.5.1 Apparatus.

a. Volumetric flask.

b. Whatman #44 and #42 filter paper, or equal.

c. 400 milliliter beaker.

d. Ashless filter paper.

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4.7.5.2 Reagents.

a. Phenolphthalcin indicator solution.

b. HN_4OH.

c. $(NH_4)_2\ C_2O_2 \cdot H_2H$.

d. $(NH_4)_2\ HPO_4$.

e. NH_4NO_3.

4.7.5.3 Procedure. The test shall be performed in accordance with the following steps:

a. Dilute the filtrate saved from 3.7.4.3, to 250 milliliters in a volumetric flask. To a 50-milliliter aliquot, add weak ammonium hydroxide solution until the pH of the solution is about 7. (Green color is formed when bromthymol blue is used as indicator.)

b. Filter the solution through a #44 Whatman filter paper. Wash the precipitate with 5 portions of 10 milliliters distilled water. Collect the filtrate and washings in a 400-milliliter beaker and heat the solution to boiling. Slowly add 20 milliliters of saturated ammonium oxalate solution while stirring the solution. Bring back to a boil while stirring, and allow to settle for 20 minutes.

c. Filter through a #42 Whatman filter paper. Wash the precipitate four times with hot water.

d. Dilute the solution to about 150 milliliters, add about 12 grams of $(NH_4)_2\ HPO_4$ dissolved in a little water and a few drops of phenolphthalein indicator solution.

e. Heat to just below the boiling point, and slowly add 1.5 N ammonia water until a faint pink color is obtained and a slight precipitation takes place. Stir well for about minute, touching the side of the beaker as little as possible. When the precipitate has become distinctly crystalline, add more ammonia until a deep color is obtained with the phenolphthalein.

f. Allow the solution to cool, add one-fifth the solution's volume of concentrated ammonium hydroxide, and allow to stand overnight.

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g. Use an ashless filter paper for filtering off the magnesium ammonium phosphate precipitate. After washing it with 1.5 N NH_4OH, moisten with a saturated solution of ammonium nigrate in 1.5 N NH_4OH, dry and ignite very slowly and carefully, and weigh. The precipitate contains 36.21 percent MgO.

h. Calculate as follows:

$$\text{Percent MgO} = \frac{\text{Wt } Mg_2P_2O_7 \times .3621}{\text{Sample Wt (W)}} \times 100$$

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified, packaging and packing shall be level C (see 6.2).

5.1.1 Level C packaging. Material shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.1.2 Level C Packing. Packages shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Rules as applicable.

5.2 Shipping containers. The material shall be shipped in commercial type containers that are suitably sealed to prevent contamination of the contents by moisture or foreign matter and shall have resealing features. Shipping containers shall be constructed to withstand normal transportation and handling abuse and shall be suitable for indoor storage for a period of one year minimum.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use as a filter for insulating materials.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Whether or not a preproduction sample is required (See 3.1).

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6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS 11034, dated 14 September 1964.

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